This is going to be a broad overview of the ideas and techniques used to prove the following much celebrated theorem.

> **Theorem** (Gromov's Theorem on Groups of Polynomial Growth)**:** If a finitely generated group Γ has polynomial growth, then Γ contains a nilpotent subgroup of finite index.

All of this work is discussed in Gromov's paper, and I am only writing this up for personal use (and to present to the University of Virginia Graduate Students Seminar).

Notions from Riemannian Geometry

> **Definition:** Let $f: X \to Y$ be a map. We call such a map *globally expanding* if, for any two points $x_1 \neq x_2$ in X, one has $d(f(x_1), f(x_2)) > d(x_1, x_2)$.
>
> We say f is expanding if each point $x \in X$ has a neighborhood U such that $f|_U$ is globally expanding.

Consider a compact connected Riemannian manifold, and let $f: X \to X$ be an expanding map. Then, X has no boundary. There is then a map $\widetilde{f}: \widetilde{X} \to \widetilde{X}$ induced on the universal cover. Since any globally expanding map is a covering map on itself, it follows that the map $\widetilde{f}$ is a homeomorphism, and the inverse map $\widetilde{f}^{-1}: \widetilde{X} \to \widetilde{X}$ is a contraction.

Furthermore, for any $\delta > 0$, there is some $\varepsilon > 0$ such that for any $y_1, y_2 \in \widetilde{X}$ with $d_{\widetilde{X}}(y_1, y_2) \geq \delta$, one has

$$d_{\widetilde{X}}\left(\widetilde{f}^{-1}(y_1), \widetilde{f}^{-1}(y_2)\right) \leq (1-\varepsilon) d_{\widetilde{X}}(y_1, y_2),$$

which follows by the definition of globally expanding. Therefore, $\widetilde{f}^{-1}$ has a unique fixed point by the contraction mapping theorem, so projecting back down yields that $f: X \to X$ has a fixed point as well.

Two theorems of Shub and Franks are fundamental to the proof. To understand them, though, we require understanding a few results.

Growth Rates and some Established Results

> **Definition:** Suppose Γ has polynomial growth. We will let growth(Γ) be the smallest natural number d such that there exists C such that $|B(r)| \leq Cr^d$ for each $r \geq 1$. Here, $B(r)$ denotes the closed unit ball of radius r centered at the identity.

Suppose $\Gamma' \leq \Gamma$ is a subgroup of a finitely generated group acting via translation. Let $X = \Gamma/\Gamma'$ be the quotient space, and let $f: \Gamma \to X$ the natural projection. Let $\text{dist}(x, y) = \inf_{\alpha,\beta} d(\alpha, \beta)$ for any $\alpha \in f^{-1}(x)$ and $\beta \in f^{-1}(y)$. Since Γ' acts via isometries on Γ, it follows that $\text{dist}(x, y)$ is a metric on X.

Observe that for any two points $x, y \in X$ with $\text{dist}(x, y) = p$ for some integer p, there exist points $x = x_0, x_1, \ldots, x_p = y$ such that $\text{dist}(x_i, x_{i-1}) = 1$. Importantly, this means that if X is infinite, then $|B_X(x, r)| \geq r + 1$ for all $x \in X$ and all $r \geq 0$.

> **Lemma** (Splitting Lemma)**:** If $\Gamma' \subseteq \Gamma$ is a *finitely generated* subgroup of *infinite* index. Then, growth(Γ') $\leq$ growth(Γ) $- 1$.

> *Proof.* Since Γ/Γ' is connected, it follows that $B(r)$ for each $r \in \Gamma$ contains at least $r+1$ elements $\alpha_0, \ldots, \alpha_r \in \Gamma$ such that $f(\alpha_i) \neq f\left(\alpha_j\right)$ whenever $i \neq j$, where $f: \Gamma \to \Gamma/\Gamma'$ is the quotient map. The intersections $B' = B(r) \cap \Gamma'$ and the translates $B'\alpha_i$ are disjoint, and they are contained in $B(2r)$, so that $|B(2r)| \geq (r+1)|B'|$. □

Linear Representations of Certain Finitely Generated Groups

> **Theorem** (Jordan's Theorem)**:** For any Lie group L with finitely many components, there is a number q such that every finite subgroup in L contains an abelian subgroup of index at most q.

Theorem (Tits Alternative)**:** Let L be a Lie group with finitely many components, and let $G \subseteq L$ be an arbitrary finitely generated subgroup. Then, either

- G has a free group of rank 2 (hence is of exponential growth);
- G is virtually nilpotent (hence G has exponential growth unless it is virtually nilpotent).

Theorem (Milnor–Wolf Theorem)**:** If Γ is a virtually solvable group, then either Γ is of exponential growth or Γ is virtually nilpotent.

Remark: Given a nilpotent group G with lower central series

$$G = \gamma_1(G) \geq \gamma_2(G) \geq \cdots \geq \gamma_c(G) \geq \gamma_{c+1}(G) = \{e_G\},$$

the Bass–Guivarch formula gives that G grows polynomially with degree given by

$$d = \sum_{i=1}^{c} i \operatorname{rk}(\gamma_i(G)/\gamma_{i+1}(G)),$$

where rk denotes the rank of the abelian groups $\gamma_i(G)/\gamma_{i+1}(G)$.

Lemma (Lemma (a))**:** Let L be a Lie group with finitely many components, and suppose that for each $p = 1, 2, \ldots$, there is a homomorphism $G \to L$ such that its image is finite and has at least p elements. Then, G contains a subgroup $G' \subseteq G$ of finite index such that $[G', G'] \subseteq G'$ has infinite index.

Proof. We start by observing that if G is a finitely generated group with finite generating set S, then given any index k, G has finitely many subgroups of index k. This follows from the fact that if H is any such subgroup, then the action of G on G/H yields a homomorphism $G \to S_k$ (and so H is determined by the given homomorphism), of which there are at most $(k!)^{|S|}$ such homomorphisms.

Therefore, there are finitely many subgroups of index at most q, and since the finite intersection of finite index subgroups is finite index, it follows that if we let

$$G' \coloneqq \bigcap \{H \leq G : [G : H] \leq q\},$$

then G' is of finite index.

We will show that $[G', G']$ is of infinite index by showing that G' admits a surjection onto abelian subgroups of arbitrary size. If $\phi \colon G \to L$ has image $\left|F_p\right| \geq p$, then $G' \subseteq \phi^{-1}\left(A_p\right)$, where $A_p \leq F_p$ is abelian and has index q. Since

$$\left[F_p : \phi(G')\right] \leq [G : G'],$$

it follows that

$$\left|\phi(G')\right| \geq \frac{p}{[G : G']}.$$

Since p is arbitrary, it follows that $[G', G'] \subseteq G'$ has infinite index. □

Lemma (Lemma (b))**:** Suppose Γ is a finitely generated group of polynomial growth. Then, $[\Gamma, \Gamma]$ is finitely generated.

Proof. It suffices to show that the kernel $\Delta \subseteq \Gamma$ of any homomorphism $g \colon \Gamma \to \mathbb{Z}$ is finitely generated.

Let $\gamma_0, \gamma_1, \ldots, \gamma_k \in \Gamma$ be a system of generators such that $g(\gamma_0) = 1 \in \mathbb{Z}$ and $\gamma_i \in \Delta$ for each $i = 1, \ldots, k$. Let

$$\Delta_m = \left\langle \left\{ \gamma_0^j \gamma_i \gamma_0^{-j} : 1 \leq i \leq k, -m \leq j \leq m \right\} \right\rangle.$$

Then,

$$\Delta = \bigcup_{m=0}^{\infty} \Delta_m.$$

If the sequence of Δ_m stabilizes, then it equals Δ and we are done. Else, there are $\alpha_m \in \Delta$ such that each α_m is of the form $\alpha_m = \gamma_0^m \gamma_i \gamma_0^{-m}$ or $\alpha_m = \gamma_0^{-m} \gamma_i \gamma_0^m$ satisfying $\alpha_m \notin \langle \alpha_0, \alpha_1, \ldots, \alpha_{m-1} \rangle$.

Were such a family to exist, our goal is to show that it implies the existence of an exponentially growing family. Let $\beta(\epsilon_0, \ldots, \epsilon_m) = \alpha_0^{\epsilon_0} \cdots \alpha_m^{\epsilon_m}$, where $\epsilon_i \in \{0, 1\}$ for each i. Since equality between any two such elements β is defined by equality of each ϵ_i, it follows that there are 2^{m+1} such β.

Notice then that $\ell(\beta) \leq \sum_{i=1}^{m} \ell(\alpha_m) \leq (m+1)(2m+1)$. Thus, $|B((m+1)(2m+1))| \leq 2^{m+1}$ as the ball of radius $(m+1)(2m+1)$ has all the β. Yet, this contradicts polynomial growth. □

The most important applications of these are to prove the following lemma.

Lemma (Lemma 1)**:** Let Γ be a finitely generated group of polynomial growth, and let L be a Lie group with finitely many connected components. Suppose that for each finitely generated infinite group $\Gamma' \subseteq \Gamma$, there is a subgroup $\Delta \subseteq \Gamma'$ of finite index such that, for all $p = 1, 2, \ldots$, there is a homomorphism $\Delta \to L$ such that its image contains at least p elements.

Then, Γ is virtually nilpotent.

Proof. By the splitting lemma, we inductively assume that all finitely generated subgroups of Γ with infinite index are virtually nilpotent.

Let $\Delta \subseteq \Gamma$ be a subgroup of finite index with the required homomorphisms into L. If these homomorphisms are all finite, Lemma (a) yields a subgroup $\Delta' \subseteq \Delta$ with finite index such that $[\Delta', \Delta'] \subseteq \Delta$ has infinite index. Induction yields the final result in this case.

If there is a homomorphism $\Delta \to L$ with infinite image, then the Tits theorem applied to this image yields a finite index subgroup $\Delta' \subseteq \Delta$ such that $[\Delta', \Delta'] \subseteq \Delta$ is infinite index as well. According to Lemma (b), the commutator $[\Delta', \Delta']$ is finitely generated, so by the induction hypothesis, it is virtually nilpotent. Therefore, Γ is virtually solvable, so by the Milnor–Wolf Theorem, it follows that Γ is virtually nilpotent. □

Constructing the Asymptotic Cone

In Gromov's original paper, he constructed a special metric space Y as a subsequential limit of the sequence of discrete metric spaces $(\Gamma, (1/n)d_S)$, where S is a generating set for S. The metric space Y has the following properties:

(i) Y is homogeneous;

(ii) Y is connected and locally connected;

(iii) Y is complete;

(iv) Y is locally compact and of finite topological dimension.

The deep result of Gleason–Montgomery–Zippin gives that $\mathrm{Isom}(Y)$ is then a connected Lie group. There is then a finite-index subgroup $\Gamma' \subseteq \Gamma$ that admits a homomorphism $\Gamma' \to \mathrm{Isom}(Y)$, and we use Lemma 1 to then obtain the desired result for Γ.

Definition: Given a metric space Y with a distinguished point e, $\mathrm{Isom}(Y)$ is topologized by defining the basic closed neighborhoods

$$U_{k,\varepsilon} = \{\sigma \in \mathrm{Isom}(Y) : k \in \mathbb{Z}_{>0} \text{ and } \varepsilon > 0 \text{ with } d(\sigma(y), y) \leq \varepsilon \text{ for all } y \text{ with } d(y, e) \leq k\}$$

at the identity.

The way we will construct this is through the asymptotic cone.

The Nonstandard Reals

Let $\mathbb{R}$ denote the real numbers, and let $\mathcal{U}$ be an ultrafilter on $\mathbb{N}$. On the set $\mathbb{R}^{\mathbb{N}}$, we may define an equivalence relation by saying two sequences $(u_n)_n$ and $(v_n)_n$ are equivalent if

$$\{n \in \mathbb{N} : u_n = v_n\} \in \mathcal{U}.$$

We denote by $\mathbb{R}^{\mathcal{U}}$, which will be written as $\mathbb{R}^*$, the ultrapower of $\mathbb{R}$. It c an be shown that the operations that turn $\mathbb{R}$ into an ordered field, such as addition, subtraction, multiplication, division, and the ordering on $\mathbb{R}$, are elementary expressions, so by the Løs theorem, they admit extensions to $\mathbb{R}^*$ that turn $\mathbb{R}^*$ into an ordered field. We will identify $\mathbb{R} \subseteq \mathbb{R}^*$ by taking $s \mapsto [s]$, where $[s]$ is the equivalence class of the constant sequence with value s.

In particular, we define

$$\begin{aligned}
[(u_n)_n] \pm [(u_n)_n] &= [(u_n \pm v_n)_n] \\
[(u_n)_n][(v_n)_n] &= [(u_n v_n)_n] \\
[(u_n)_n]/[(v_n)_n] &= \begin{cases} u_n/v_n & v_n \neq 0 \\ 0 & v_n = 0, \end{cases}
\end{aligned}$$

and we say that $[(u_n)_n] \leq [(v_n)_n]$ if and only if $\{n : u_n \leq v_n\} \in \mathcal{U}$.

We note that the nonstandard construction does not preserve the Archimedean property. In particular, we may define an element $[u]$ to be *finite* if there exists $s \in \mathbb{R}$ with $s > 0$ such that $[-s] \leq [u] \leq [s]$, and we say the element is *infinite* if it is not finite. We say the element is *infinitesimal* if for all $s \in \mathbb{R}$ with $s > 0$, we have $[-s] \leq [u] \leq [s]$. If $\mathbb{R}^{\text{fin}}$ denotes the finite elements of $\mathbb{R}^*$ and $\mathbb{R}^0$ denotes the infinitesimal elements, then there is an isomorphism $\rho\colon \mathbb{R} \to \mathbb{R}^{\text{fin}}/\mathbb{R}^0$; if $h\colon \mathbb{R} \to \mathbb{R}^{\text{fin}}$ is the natural inclusion map, then the homomorphism $\text{st} = \rho^{-1} \circ h$ from $\mathbb{R}^{\text{fin}}$ to $\mathbb{R}^0$ is called the *standard part* map.

Constructing the Space Y

> **Definition:** Let ω be an ultrafilter on $\mathbb{N}$, and let Γ be a finitely generated group. We will let $\Gamma^{(M)}$ denote the space of sequences

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